Synoptic Question 6: The Physics of a Professional Tennis Serve

Reading Passage

Read the following article and then answer the questions that follow.

The tennis serve is one of the most powerful and complex motions in sports. It is a sequence of coordinated movements that transfers energy from the player's body into the racquet and finally into the ball, launching it at high speed. A professional player can strike the ball at speeds exceeding 220 km/h.

The effectiveness of the serve depends on maximizing this final ball speed while maintaining control over its trajectory. The ball must be struck at a sufficient height, travel over the net, and land within the designated service box on the opposite court.

The physics of the impact itself is fascinating. The ball, which is momentarily at rest at the peak of the toss, is struck by the racquet head. During the very short impact time (on the order of milliseconds), the ball is compressed, and the elastic strings of the racquet stretch and then rebound, launching the ball. This collision is an example of an inelastic collision, where kinetic energy is not perfectly conserved, but momentum is. The distinctive "ping" sound heard is a shock wave produced by the rapid compression and expansion of the ball and strings.

Engineers design racquets to maximize power and control. This includes optimizing the racquet's mass, balance, and the tension of its strings. The "sweet spot" of the racquet is a point of impact that maximizes the power transferred to the ball while minimizing the shock and vibration transmitted to the player's arm.

Data:

A professional player hits a serve. The following data is recorded:

Table 1: Data for a Tennis Serve

Parameter	**Value**
Mass of tennis ball, m	0.057 kg
Speed of ball just after impact, v_{ball}	220 km/h
Height of ball at impact, h_0	2.70 m
Horizontal distance from player to net, d_{net}	11.89 m
Height of the net, h_{net}	0.914 m
Distance from net to service line, d_{box}	6.40 m
Contact time between ball and racquet, Δt	5.0 ms
Speed of sound in air, v_{sound}	343 m s^{-1}
Frequency of sound from impact, f	520 Hz

Questions

(a) Convert the speed of the ball just after impact, v_{ball}, to m s^{-1}.

(b) Assuming the ball is at rest at the peak of its toss when struck, calculate:

  (i) The change in momentum (impulse) delivered to the ball.

  (ii) The average force exerted by the racquet on the ball during the impact.

(c) Calculate the kinetic energy of the ball just after it leaves the racquet.

(d) The work done by the racquet on the ball is equal to the ball's final kinetic energy. Calculate the average power delivered to the ball during the impact.

(e) For this serve, the ball is hit horizontally (at an angle of 0° to the horizontal). For the purpose of calculation, ignore the effects of air resistance and spin.

  (i) Calculate the time it takes for the ball to travel the horizontal distance to the net.

  (ii) Calculate the vertical distance the ball drops due to gravity during this time.

  (iii) By finding the ball's height as it passes the net, determine if the serve is successful (i.e., if it clears the net).

(f) Using your calculations from (e):

(i) Calculate the total horizontal distance the ball travels before it hits the ground.

(ii) Determine if the ball lands inside the service box. (The service box extends from the net to 6.40 m beyond the net).

(g) (i) Calculate the wavelength of the sound wave produced by the impact.

(ii) How much time passes between the player hitting the ball and a spectator hearing the sound from a distance of 15.0 m?

Solutions

(a) $v_{ball} = 220 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 61.11 \text{ m s}^{-1}$.

(b) (i) $\Delta t = 5.0 \text{ ms} = 0.0050 \text{ s}$.
Impulse $= \Delta p = m(v_f - v_i) = (0.057)(61.11 - 0) = 3.48$ Ns.

(ii) Average Force, $F_{avg} = \frac{\Delta p}{\Delta t} = \frac{3.48}{0.0050} = 696$ N.

(c) $KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.057)(61.11)^2 = 106.3$ J.

(d) Average Power, $P_{avg} = \frac{\text{Work Done}}{\text{time}} = \frac{KE}{\Delta t} = \frac{106.3}{0.0050} = 21260$ W (or 21.3 kW).

(e) This is a projectile motion problem with $u_x = 61.11 \text{ m s}^{-1}$ and $u_y = 0 \text{ m s}^{-1}$.

(i) Time to net, $t_{net} = \frac{d_{net}}{u_x} = \frac{11.89}{61.11} = 0.1946$ s.

(ii) Vertical drop, $s_y = u_y t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(9.81)(0.1946)^2 = 0.1857$ m.

(iii) Height at net $= h_0 - s_y = 2.70 - 0.1857 = 2.51$ m.
Since 2.51 m $>$ 0.914 m (the net height), the serve successfully clears the net.

(f) (i) First, find the total time for the ball to drop the full height of 2.70 m.
$s_y = u_y t + \frac{1}{2}a_y t^2 \implies 2.70 = 0 + \frac{1}{2}(9.81)t_{total}^2$
$t_{total} = \sqrt{\frac{2 \times 2.70}{9.81}} = 0.741$ s.
Total horizontal distance, $d_{total} = u_x \times t_{total} = 61.11 \times 0.741 = 45.28$ m.

(ii) The service box ends at $d_{net} + d_{box} = 11.89 + 6.40 = 18.29$ m.
The ball lands at 45.28 m, which is far beyond the service box. The serve is "out".
(Note: This is why players must hit the ball downwards at a significant angle, not horizontally).

(g) (i) Wavelength, $\lambda = \frac{v_{sound}}{f} = \frac{343}{520} = 0.660$ m.

(ii) This is the time for the sound wave to travel 15.0 m.
Time $= \frac{\text{distance}}{\text{speed}} = \frac{15.0}{343} = 0.0437$ s.

[Diagram showing the trajectory of a tennis serve over the net]

Figure 1: Trajectory of a Tennis Serve (not to scale)

Cambridge International AS & A Level Physics (9702)

Paper 4: A Level Structured Questions
Synoptic Question 6 – Variant 2 (Higher Difficulty)

Theme: The Physics of a Professional Tennis Serve (Advanced Modelling)

Total Marks: 30 **Time guidance:** 45 minutes

Instructions:

- Answer all parts.
- Give your working and appropriate units.
- Use $g = 9.81\,\text{m s}^{-2}$ and speed of sound in air $= 343\,\text{m s}^{-1}$ unless otherwise stated.

Question 6 Variant 2

Reading passage (extract the relevant numerical data before answering):

In a training study on first-serve mechanics, a player strikes a tennis ball of mass 5.7×10^{-2} kg. Video analysis shows that immediately after impact the ball speed is $58.0\,\text{m s}^{-1}$, directed at $7.0°$ below the horizontal, from a contact height of 2.65 m above court level. The net is located 11.89 m from the server and is 0.914 m high. For a legal serve, the ball must land in the service box, which extends 6.40 m beyond the net. During impact the contact duration is measured as 4.2 ms, while the incoming ball speed just before contact is $8.0\,\text{m s}^{-1}$ opposite to the outgoing direction.

For modelling, assume air resistance is negligible in parts (a)–(d), and use $F_D = kv_x$ with $k = 1.8 \times 10^{-3}\,\text{N s m}^{-1}$ for part (e).

(a) Resolve the launch velocity into horizontal and vertical components, taking upward as positive. [2]

(b) Determine:

  (i) the time taken for the ball to reach the net,

  (ii) the height of the ball above the court as it passes the net,

  (iii) whether the ball clears the net and by how much.

[6]

(c) Calculate the horizontal distance from the server at which the ball first lands. Hence decide whether the serve lands in the service box. [4]

(d) During impact, contact time is 4.2 ms and the ball approaches the racquet at $8.0\,\text{m s}^{-1}$ in the opposite direction to the outgoing ball. Estimate:

  (i) the impulse delivered to the ball,

  (ii) the average force exerted by the racquet on the ball,

(iii) the percentage increase in kinetic energy of the ball.

[8]

(e) **Open-ended modelling and evaluation.**
A more realistic model uses a horizontal drag force $F_D = kv_x$ with $k = 1.8 \times 10^{-3}\,\mathrm{N\ s\ m^{-1}}$, acting opposite to motion.
The racquet is restrung at different tensions, changing effective contact stiffness and dwell time.
Discuss, with clear physics reasoning, how including drag, topspin (Magnus effect), and string tension changes:

(i) time of flight,

(ii) horizontal range,

(iii) net-clearance margin,

(iv) tactical value of the serve,

(v) the trade-off between control and ball speed for higher vs lower string tension.

Your answer should include appropriate equations, assumptions, and at least one justified limitation of your model. [10]

Sample Solution (Variant 2)

Given: $m = 0.057\,\text{kg}$, $v = 58.0\,\text{m s}^{-1}$, $\theta = 7.0^\circ$ below horizontal, $y_0 = 2.65\,\text{m}$.

(a)

$$v_x = v\cos 7.0^\circ = 57.57\,\text{m s}^{-1}$$
$$v_y = -v\sin 7.0^\circ = -7.07\,\text{m s}^{-1}$$

(b)

$$t_{\text{net}} = \frac{11.89}{57.57} = 0.2065\,\text{s}$$

Vertical position at net:

$$y = y_0 + v_y t - \tfrac{1}{2}gt^2 = 2.65 + (-7.07)(0.2065) - \tfrac{1}{2}(9.81)(0.2065)^2 = 0.98\,\text{m}$$

Net clearance:

$$0.98 - 0.914 = 0.066\,\text{m}$$

So the ball clears the net by about 6.6 cm.

(c) Set $y = 0$:

$$0 = 2.65 - 7.07t - 4.905t^2$$
$$t = 0.288\,\text{s} \quad \text{(positive root)}$$

Horizontal distance:

$$x = v_x t = 57.57(0.288) = 16.6\,\text{m}$$

Service line limit from server:

$$11.89 + 6.40 = 18.29\,\text{m}$$

Since $16.6 < 18.29$, the serve lands **in** the service box.

(d) Incoming velocity is opposite to outgoing direction, so with chosen positive direction:

$$u = -8.0\,\text{m s}^{-1}, \quad v = +58.0\,\text{m s}^{-1}$$

Impulse:

$$J = m(v - u) = 0.057(58 - (-8)) = 3.76\,\text{N s}$$

Average force:

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{3.76}{4.2 \times 10^{-3}} = 8.95 \times 10^2\,\text{N}$$

Kinetic energies:

$$KE_i = \tfrac{1}{2}mu^2 = \tfrac{1}{2}(0.057)(8.0)^2 = 1.82\,\text{J}$$
$$KE_f = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(0.057)(58.0)^2 = 95.9\,\text{J}$$
$$\%\ \text{increase} = \frac{KE_f - KE_i}{KE_i} \times 100 = \frac{95.9 - 1.82}{1.82} \times 100 = 5.17 \times 10^3\% \ (\approx 5170\%)$$

(e) Open-ended discussion (example of a high-level response)

With linear drag, horizontal equation of motion becomes

$$m\frac{dv_x}{dt} = -kv_x \Rightarrow v_x = v_{x0}e^{-kt/m}, \quad x = \frac{mv_{x0}}{k}\left(1 - e^{-kt/m}\right).$$

Hence v_x decreases with time, so horizontal range is reduced compared with no-drag motion.

Time of flight is affected mainly through coupled vertical dynamics if full drag is included. Even with horizontal-only drag, reduced forward speed means the ball reaches the net slightly later; this increases downward displacement due to gravity before the net, reducing clearance margin.

Topspin introduces a downward Magnus force, increasing effective downward acceleration. This shortens flight arc, can improve control, and helps powerful serves dip into the service box. The trade-off is that excessive topspin may reduce peak speed and make the serve less penetrating.

String tension changes impact behaviour. A higher string tension generally gives a stiffer string bed and shorter dwell time, so less trampoline effect and usually lower rebound speed for the same swing; however, deformation is smaller and directional control is often improved. Lower tension typically increases dwell time and effective energy return, which can raise outgoing speed, but launch-angle variation may increase and control can reduce for the same stroke mechanics.

Tactically, players balance speed, spin, placement, and tension setup: high speed pressures return time, while spin and higher-tension control support accuracy and second-serve reliability.

One model limitation: taking $F_D \propto v$ is a simplification; at tennis-ball Reynolds numbers drag is closer to a mixed/nonlinear dependence and depends on seam orientation and spin rate. Therefore precise trajectory prediction needs measured aerodynamic coefficients and 2D/3D numerical integration.

Mark Scheme (CIE P4 style, Variant 2)

Total: 30 marks

Part	Expected answer points	Marks	AO
(a)	$v_x = 57.6\,\mathrm{m\ s^{-1}}$ and $v_y = -7.1\,\mathrm{m\ s^{-1}}$ (allow ecf, sign must be consistent with angle below horizontal).	2	AO1
(b)(i)	Correct use of $t = x/v_x$, $t \approx 0.206\,\mathrm{s}$.	2	AO2
(b)(ii)	Correct vertical equation and substitution, $y \approx 0.98\,\mathrm{m}$.	2	AO2
(b)(iii)	Compares with $0.914\,\mathrm{m}$ and concludes clears net by $\approx 0.07\,\mathrm{m}$.	2	AO2
(c)	Solves quadratic for landing time (positive root) and uses $x = v_x t$ to get $x \approx 16.6\,\mathrm{m}$; valid in/out decision versus $18.29\,\mathrm{m}$.	4	AO2
(d)(i)	Correct momentum change/impulse including opposite direction, $J \approx 3.76\,\mathrm{N\ s}$.	3	AO2
(d)(ii)	$F = J/\Delta t$ with ms to s conversion, $\approx 8.9 \times 10^2\,\mathrm{N}$.	2	AO2
(d)(iii)	Correct kinetic-energy calculations and percentage increase, large value ($\sim 5.2 \times 10^3\%$).	3	AO2
(e)	Award up to 10 marks using levelled descriptors below.	10	AO3

Part (e) level descriptors (10 marks)

Level	Descriptor	Mark range
L3	Coherent, logically structured physics argument. Uses relevant equations (e.g. drag differential equation and/or Magnus-force discussion), explains how string tension alters force transfer / rebound speed and control, links changes to trajectory and tactics, and states at least one justified modelling limitation. Minor errors only.	8–10
L2	Sound explanation of most effects (range, clearance, control, tension trade-off) with partial quantitative support. Some links or assumptions underdeveloped; limitation may be generic.	5–7
L1	Basic qualitative statements (e.g. drag slows ball, spin makes ball dip, tension affects feel/control) with limited physics development and little/no model justification.	2–4
L0	No creditworthy physics.	0

Indicative content for part (e) (credit any valid physics):

- v_x decreases with time under drag, so range decreases.
- Later arrival at net can reduce clearance due to greater gravitational drop.
- Topspin adds downward Magnus force, increasing dip and in-court probability.
- Tactical trade-off between speed, spin, and placement/consistency.

- String tension trade-off: higher tension usually improves directional control but may reduce rebound speed; lower tension often gives more speed/power but less control.
- Explicit assumptions and at least one realistic limitation of the model.